ASHWATH RANJITH

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Education

- **SIES Graduate School of Technology, Navi Mumbai, India** **2023- 2027**
  Branch: Computer Engineering | CGPA: 9.95 (Overall), 10.00 (Last Semester)
- **St. Joseph's High School and Junior College, Navi Mumbai, India** **2022 - 2023**
  Stream: Computer Science | HSC (Class XII), Percentage: 93.33%

Skills

- **Languages:** C, C++, Python, Java, JavaScript, C#
- **Systems & Networking:** TCP sockets, client–server architecture, real-time systems
- **Artificial Intelligence:** NLP, OCR pipelines, applied computer vision
- **Frameworks & Platforms:** Unity, React, React Native, Flutter
- **Databases & Backend:** PostgreSQL, MySQL, Firebase, REST APIs

Projects

- **Galactic War – Mobile Space Shooter (Unity)** [Google Play Link]
  - Shipped an arcade space shooter on **Google Play** with 20 levels, an endless survival mode, and **4 boss battles**, **8 distinct enemy types**, each demanding a different strategy.
  - Engineered game systems including weapon upgrades, ship repairs, and in-game economy alongside **AdMob monetisation**.

- **Microprocessor Simulator –** Custom ISA & Assembly Interpreter (C++) [GitHub Link]
  - Designed a software microprocessor in C++ with modular components — registers, ALU, decoder, flags, and memory — supporting a **custom 12 instruction ISA.**
  - Users input raw opcodes directly, mimicking 8085 trainer-kit interaction, executed through a **custom fetch-decode-execute pipeline.**

- **Question Paper Analyzer – OCR & ML Analysis Engine** [Web Link]
  - Full-stack tool where users upload scanned exam papers; an **OCR + NLP** pipeline extracts text and applies **DBSCAN clustering** to group question variations by topic.
  - Surfaces **topic frequency charts**, **clustered question variants**, and a wildcard section for uncategorised but previously asked questions.

- **VectorForge – Web Pixel Art Editor** [Web Link]
  - Browser-based pixel art editor in vanilla HTML/CSS/JS with layers, zoom/pan, custom palettes, PNG export, and localStorage persistence across grids beyond 64×64.
  - **Vectorize feature** upscales pixel art 4× and applies **vertex identification and Bézier curve** fitting to smooth jagged edges into clean vector-style curves.

Academic and Extracurricular Achievements

- Runner-up – **SIES Cognition GameJam Hackathon** (2025): Awarded **₹5,000 cash prize** and certificate
- **R. Govindraj Scholarship (First Year, 2024)** and **Essar Endowment (Second Year, 2025)**: Awarded for highest academic performance in college
- **Highest Academic Performance (Class XII, 2023)**: Awarded shield and certificate for top academic achievement